Bahria University Lahore Campus

(Department of Computer Sciences)

Course Code/number	GSC-221
Course Title/Name	**Discrete Mathematics**
Credit Hours/Contact hours	3/ 3 hours/week
Degree Program	BSCS
Prerequisites or Co-requisites	None
Assessment Methods and Weightage	Quizzes 10 Assignments/Class Participation/Project 20 Mid-Term Examination 20 Final Examination 50 **Total** **100**
Textbook (or **Laboratory Manual** for Laboratory Courses)	1. Susanna S. Epp(2010) Discrete Mathematics with Applications, 4th Edition
Reference Material	1. *Discrete Mathematics and Its Applications*, 7th edition; by Rosen; McGraw-Hill; 2. *Discrete Mathematics by Richard Johnsonbaugh*, Prentice Hall, 0135182425. 3. *Discrete Mathematical Structures*, 4th Edition, by Kolman, Busby & Ross, 2000, Prentice-Hall.
Web Resources	
Instructor Name/Cluster Head/Subject Expert	Instructor Name: Tahir Iqbal, Designation: Snr. Assistant Professor Status: Regular Cluster Head Name: Dr. Abdul Hafeez Subject Expert Name: Ms Munaza Sher
Course Aims	Introduces the foundations of discrete mathematics as they apply to Computer Science, focusing on providing a solid theoretical foundation for further work. Further, this course aims to develop understanding and appreciation of the finite nature inherent in most Computer Science problems and structures through

	study of Graphs and trees., Elements of Set Theory, Relations, The Logic of compound Statements, Elements of Number Theory, Recursion, Mathematical Induction and pigeonhole principle.
Course Objectives	1. To provide students with a solid understanding of Discrete Mathematics. 2. To introduce students to the fundamental as well as advanced concepts of Set theory, Algorithms, Functions, Mathematical reasoning, Graph Theory, and Trees. 3. To enable students, improve problem-solving skills of enumerating objects using combinatorial analysis. 4. To enable students, learn about the abstract Mathematical structures used to represent 5. discrete objects and relationships between these objects

Course Outcomes	**After successful completion of this course, the students should be able to:**	**PLO**	**BT Level**
	1. **Demonstrate** an understanding of the key concepts of Discrete Structures. 2. **Relate** various discrete structures with different areas of Computer Science. 3. **Apply** formal logic proofs and/or informal, but rigorous, logical reasoning to real world problems. 4. **Develop** pseudocodes of algorithms for standard computing problems.	1 2 3 4	C2 C2 C3 C3

Course Description/Catalogue	Trees and Graphs, sets, logic, Number theory, Recursion, Mathematical Induction, pigeonhole principle.

Lecture Plan (16 Weeks)	Week # and Date	Lecture /Hour #	Topic to be covered	Learning outcomes	Reference Text
	1	1	**Mathematical Logics I** Introduction, What is Discrete Mathematics? Logic, simple statements, compound statements	To know about course objective, introduction to logics	Susanna Ch-2
		2	Negation, Conjunction, Disjunction. Truth tables, De Morgan's Laws, Tautology	To know about truth table representation , demorgan's law and Law of tautology	Susanna Ch-2

	2	1.	**Mathematical Logics II** Applying laws of logics, Conditional statements, Negation of conditional statement, inverse	Solve conditional statements, negations, and inverse	Susanna Ch-2
		2.	Converse and contra-positive of conditional statements. Bi-conditional, Laws of logic	To learn more about conditional statements and various laws of logics	Susanna Ch-2
	3	1.	**Mathematical Logics III** Argument, Valid and Invalid argument.	To identify valid and invalid arguments	Susanna Ch-2
		2.	Switches in series, switches in parallel, AND-gate, OR-gate, Combinational circuit, Determining output.	To relate logics to real life situations	Susanna Ch-2
	4		**Sets** Introduction of sets, Tabular, Descriptive and Set builder forms of set, subset, proper subset, equal set, null set	To introduce concept of set	Susanna Ch-6
			Universal set, Venn diagram, Finite and Infinite set, properties of sets.	Representation of situations using Venn Diagram	Susanna Ch-6
	5	1.	**Sets** Set identities, Power set	Learn Set identities and power set.	Susanna Ch-6
		2.	De Morgan's Law, Venn diagram in set identities	To solve real life problems using Venn diagram	Susanna Ch-6
	6	1	**Relations** Binary Relations, inverse relation, domain of relation Representation of relations.	To introduce concept of Binary Relations	Susanna Ch-8

		2	Properties of relation, types of relation, equivalence relations	Understand Propertis and Types of relations	Susanna Ch-8
	7	1	**Functions I** Function and its representation, Algebra of functions	Understand functions	Susanna Ch-7
		2	Composition functions	Understand composition of functions	Susanna Ch-7
	8	1.	**Functions II** One to one, onto and bijective functions and their graphs	Understand types functions	Susanna Ch-7
		2.	Inverse of a function	Understand inverse of functions	Susanna Ch-7
			Mid Term		
	9	1.	**Sum and Sequences** Sequence, Arithmetic and Geometric sequence,	Solve problems related to arithmetic and geometric sequences	Susanna Ch-5
		2.	Series, Arithmetic and Geometric series, Infinite geometric series.	Solve problems related to arithmetic and geometric series	Susanna Ch-5
	10	1	**Recursion** Recursion, Recursively defined function, Use of recursion.	Understanding of recursively defined functions	Susanna Ch-5
		2	Recursively defined set, Recursive definition of sum, Recursive definition of union of sets, Recursive definition of intersection of sets.	Understanding of recursive definitions	Susanna Ch-5
	11	1.	**Mathematical Inductions** Principal of mathematical induction.	Prove equations using Mathematical induction	Susanna Ch-5
		2.	Mathematical induction for divisibility problems and inequality problems	Prove equations using Mathematical induction	Susanna Ch-5

	12	1.	**Graph Theory I** Paths and Circuits.	To know basics of graph theory	Rozen Ch-4
		2.	Matrix representation of graphs	Learn how to represent graphs in Matrix	Rozen Ch-8
	13	1.	**Graph Theory II** Isomorphism of graphs Planar graph	Understand Isomorphism and planar graph	Susanna Ch-10
		2	Euler's Formula, Graph coloring.	Understan Euler's formua	Susanna Ch-10
	14	1.	**Trees** Tree, Properties of tree.	To learn properties of trees	Susanna Ch-10
		2.	Rooted tree, binary tree.	To understand what is rooted and binary tree	Susanna Ch-10
	15	1.	**Spanning trees** Matrix-tree theorem, weighted graph,	To get student prepare for problems related to Minimum spanning tree algorithms	
		2.	Minimal spanning trees,	Practice of Kruskal's Algorithm and Prim's algorithm	
	16	1.	Kruskal's Algorithm and Prim's algorithm		
		2.	**Revision**		

Assignments/ Projects and quizzes Plan	Week #	Assignment No.	Quiz No.	Project Title	Assignment/Project Quiz Date	Result Date of Assignment/Project/Quiz

	1					
	2					
	3		Quiz-1			
	4				Quiz-1	
	5	Assignment-1				Quiz-1
	6				Assignement-1	
	7					
	8	Assignment-2	Quiz-2			Assignement-1
	10					
	11		Quiz-3		Assignment-2	
	12	Assignment-3				Assignment-2
	13		Quiz-4		Assignment-3	
	14					Assignment-3

PLOs/CLOs	CLO1	CLO2	CLO3	CLO4
PLO1 (Academic Education)	✓			
PLO2 (Knowledge for Solving Computing Problems)		✓		
PLO3 (Problem Analysis)			✓	
PLO4 (Design/Development of Solutions)				✓
PLO5 (Modern Tool Usage)				
PLO6 (Individual and Teamwork)				
PLO7 (Communication)				
PLO8 (Computing Professionalism and Society)				
PLO9 (Ethics)				
PLO10 (Life-long Learning)				

Updated by: TAHIR IQBAL **Date:**